BANKING TECHNOLOGY LAB: FD CALCULATOR

Comprehensive Code-Level Audit & Compliance Verification Report

Microservice Architecture | Spring Boot 3.3.4 | Java 21 | JUnit 5 | OAuth2 JWT Resource Server
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SPECIAL. EXTERNAL DATABASE SCHEMA VERIFICATION

An external database table screenshot was provided showing columns: created_at, updated_at, ldbid INT, ooo_acpt_ts_utc_ofst VARCHAR(6), ooo_user_id VARCHAR(32), ooo_ws_id VARCHAR(32), rule_system_id VARCHAR(32), uuid VARCHAR(36), ooo_acpt_ts DATETIME(6), ooo_host_ts DATETIME(6), ooo_local_ts DATETIME(6).

Inspection Question	Findings & Code Evidence	Status
1. Does this exact table exist?	Searched entire repository, SQL scripts, resources. No matches.	'L NOT FOUND
2. Does any FD Calculator table contain uuid VARCHAR(36)?	Only table is fd_calculation_audit (init.sql / CalculationAudit.java). No uuid column.	'L NO UUID IN DB
3. Does any other table contain uuid VARCHAR(36)?	Zero other database tables exist in this microservice.	'L ABSENT
4. What is the table in the screenshot?	Belongs to an external enterprise order-orchestration / trading core (ooo_ = Order Orchestration / Out of Order, ldbid = Logical Database ID, rule_system_id = Rule Engine ID).	EXTERNAL SCHEMA
5. What is the purpose of UUID in project?	UUID is generated strictly in-memory by CorrelationIdFilter.java for X-Request-ID and SLF4J MDC. Never stored in SQL.	EPHEMERAL TRACE
6. Is it related to calculation_id or fd_account_no?	calculation_id uses CALC-YYYYMMDD-0001 (sequence generator). fd_account_no uses branch+seq+check (L12). Neither is a UUID.	UNRELATED

1. EXECUTIVE SUMMARY & READINESS SCORE

OVERALL VERDICT: READY WITH MINOR GAPS (Academic Viva Risk: MODERATE TO HIGH)

The FD Calculator microservice is a clean, robust, production-grade Spring Boot 3.3.4 implementation with 52/52 passing tests, pure BigDecimal math, and dual JWT security.
However, 4 explicit PPT Slide 7 requirements are missing or partial in actual code: APY is not implemented, Start Date input is absent, Periodic Payout is validated but not calculated, and Rate Caching is 100% missing.

Evaluation Dimension	Score	Actual Implementation Status Summary
Core Mathematical Engine	95%	Flawless BigDecimal simple and compound interest with pure Taylor series ln/exp exponentiation.
Security & Auth Integration	95%	OAuth2 Resource Server with JWT signature verification, role enforcement, and customer data isolation.
Lab Manual L6, L10, L11 Compliance	85%	Satisfies bonus calculations, bonus caps, product caps, and error handling. Uses Java CSV over Python.
Lab PPT Slide 7 Compliance	65%	Lacks APY calculation, Start Date parameter, Rate Caching, and periodic payout schedule calculation.
Automated Test Coverage	100%	52 automated tests (15 math engine, 10 bonus rules, 13 service orchestration, 6 controller, 8 WireMock).

2. PPT REQUIREMENTS AUDIT (SLIDE 7 - DR. PRAKASH K. AITHAL)

Slide 7 of the course presentation specifies the requirements for the FD Calculator module. Every requirement was audited against the active Java code:

PPT Requirement	Status	Code Evidence / Implementation Analysis
Bank user function	Ø=ßá PARTIAL	SecurityConfig.java:L76 restricts endpoints strictly to hasRole("CUSTOMER"). Bank officers/ admins receive 403 Forbidden.
Input: Amount / Principal	' DONE	CalculationRequest.java:L27: BigDecimal principal validated with @NotNull, @DecimalMin("0.01"), @Digits.

PPT Requirement	Status	Code Evidence / Implementation Analysis
Input: Tenure (Term)	' DONE	CalculationRequest.java:L32: TermDto term supports MONTHS and YEARS, normalized to months.
Input: Interest rate	Ø=ßá PARTIAL	User cannot input rate; fetched securely from Product Service via productCode to prevent tampering.
Input: Start date	'L MISSING	CalculationRequest has no start date field. Client cannot supply start date.
Calculate: Maturity amount	' DONE	CalculationEngineImpl.java: calculateSimple and calculateCompound.
Calculate: Maturity date	Ø=ßá PARTIAL	FdCalculatorServiceImpl.java:L141: LocalDate.now().plusMonths(termMonths). Always uses current date.
Calculation methods (Simple & Compound)	' DONE	CalculationEngineImpl.java implements both with 12 decimal places intermediate precision.
Compounding frequencies	' DONE	CompoundingFrequency enum supports MONTHLY (12), QUARTERLY (4), HALF_YEARLY (2), YEARLY (1).
Interest payout on regular basis	Ø=ßá PARTIAL	PayoutMode.PERIODIC is validated against product, but calculation engine executes standard cumulative formula.
Calculate APY	'L MISSING	Zero code found for APY or yield formula. No apy field in CalculationResponse.
Product module integration	' DONE	ProductClient.java Feign client calls /internal/products/{code}/calculator-config + MockProductClient in dev.
Rate information caching	'L MISSING	Zero caching annotations (@Cacheable), zero cache configuration, spring-boot-starter-cache absent from pom.xml.
Retain caching till logout	'L MISSING	No session-level or token-scoped rate caching or logout invalidation implemented.

3. LAB MANUAL COMPLIANCE (LABS L6, L10, L11)

Lab & Requirement	Status	Actual Code Findings & Evaluation
L6: Simple & Compound Formulas	' DONE	Formulas M = P + (P*r*t/100) and M = P*(1+r/n)^(nt) exact in CalculationEngineImpl.java.
L6: Customer Category Bonuses	' DONE	CategoryBonusProperties.java: Senior Citizen (+1.0%), Employee (+0.5%), Premium (+1.0%).
L6: Combined Bonus Cap (2.0%)	' DONE	CategoryBonusServiceImpl.java:L69: requestedTotal.min(maxTotalBonus). Enforces 2.0% cap.
L6: Product Max Rate Cap	' DONE	CategoryBonusServiceImpl.java:L77: effectiveRate capped at product maxRate; sets rateCapped=true.
L6: API Endpoint & Response	Ø=ßá PARTIAL	Endpoint is /api/v1/fd/calculations (manual said /api/fd/calculate). Response contains all required fields.
L6/L11: Python Report Generation	Ø=ßá PARTIAL	Manual calls Runtime.getRuntime().exec("python3 generate_fd_report.py"). Code uses pure Java ExportService.java.
L10: JWT Authentication Required	' DONE	SecurityConfig.java requires Bearer JWT; unauthenticated calls return 401 Unauthorized.
L10: CUSTOMER Role Restriction	' DONE	SecurityConfig.java: requestMatchers("/api/v1/**").hasRole("CUSTOMER"). Wrong role returns 403.
L10: Authenticated Customer Identity	' DONE	FdCalculatorServiceImpl.java:L308: jwt.getSubject() used. Username never read from request body.
L10: Authoritative Category Retrieval	' DONE	CustomerClient.java Feign call fetches categories from Customer Service. Categories not accepted in request.
L11: Rate Limit & Boundary Tests	' DONE	Covered in CalculationEngineImplTest and CategoryBonusServiceImplTest (decimal principals, large values).
L11: Outage Handling (HTTP 503)	' DONE	GlobalExceptionHandler maps Feign failure to ServiceUnavailableException (HTTP 503 SERVICE_UNAVAILABLE).

4. REPORTING AUDIT: PURE JAVA VS PYTHON

Lab L11 specifies report generation using a shell call from Java: Runtime.getRuntime().exec("python3 generate_fd_report.py").

ARCHITECTURAL DIVERGENCE ANALYSIS

1. Actual Code Evidence: No Python scripts exist in the repository. ExportService.java generates RFC 4180 compliant CSV using Java StringWriter.
2. Security Evaluation: In production banking systems, invoking external OS processes via Runtime.getRuntime().exec() from HTTP request threads is a high-severity security risk (command injection, process starvation, environment leakage). Pure Java CSV streaming is technically superior.
3. Viva Presentation: Explain to the evaluator that the requirement was satisfied using native Java streaming to eliminate OS

dependencies and command injection vulnerabilities.

5. MATHEMATICAL CALCULATION ENGINE AUDIT

Deterministic BigDecimal Arithmetic Flow (CalculationEngineImpl.java)

All monetary math uses BigDecimal exclusively with MathContext(20, HALF_UP) and 12-decimal intermediate precision. Output amounts are strictly scaled to 2 decimal places with HALF_UP rounding.

Pure BigDecimal Exponentiation: Standard BigDecimal.pow(int) only accepts integers. To compute fractional exponents (such as 6 months at quarterly compounding: exponent = 4 * 0.5 = 2.0), the engine implements Taylor series expansions:

- Base to Exponent Identity: base^exponent = exp(exponent * ln(base))
- ln(x) = 2 * arctanh((x - 1) / (x + 1)) = 2 * (y + y^3/3 + y^5/5 + ...)
- exp(x) = 1 + x + x^2/2! + x^3/3! + ...

Zero double or float primitives are used, eliminating IEEE-754 floating-point inaccuracies.

6. GAP ANALYSIS: APY & PERIODIC PAYOUT

1. APY (Annual Percentage Yield) - 'L NOT IMPLEMENTED

The codebase was searched for APY, Annual Percentage Yield, and annualized yield formulas. Exactly 0 occurrences were found. The standard banking formula:

APY = (1 + r / (100 * n))^n - 1 (expressed as a percentage)

is not implemented in CalculationEngineImpl, and no apy field exists in CalculationResponse.java.

2. Periodic Interest Payout - Ø=ßá PARTIALLY IMPLEMENTED (LOGICAL STUB)

PayoutMode enum contains CUMULATIVE and PERIODIC. FdCalculatorServiceImpl validates whether the product supports periodic payouts. However, in lines 130-137:

if (interestType == InterestType.SIMPLE) calculateSimple(...) else calculateCompound(...);

The engine executes the exact same cumulative formula regardless of payoutMode. It does NOT compute periodic payout installments (e.g. ¹625/month) or deduce payouts from maturity amounts.

7. DATABASE & AUDIT PERSISTENCE AUDIT

Mapped by CalculationAudit.java and created via src/main/resources/db/init.sql:

Column Name	SQL Type	Java Field	Constraint / Index	Purpose
calculation_id	VARCHAR(30)	String	PRIMARY KEY	Unique simulation ID (CALC-YYYYMMDD-0001)
customer_id	VARCHAR(50)	String	INDEX (idx_customer_id)	Customer username from JWT sub claim
product_code	VARCHAR(20)	String	NOT NULL	FD product code
principal	DECIMAL(18,2)	BigDecimal	NOT NULL	Principal amount in INR
term_months	INT	int	NOT NULL	Tenure in months
base_rate	DECIMAL(6,2)	BigDecimal	NOT NULL	Base rate from product (%)
effective_rate	DECIMAL(6,2)	BigDecimal	NOT NULL	Effective rate after bonuses & cap (%)
interest_type	VARCHAR(10)	String	NOT NULL	SIMPLE or COMPOUND
interest_earned	DECIMAL(18,2)	BigDecimal	NOT NULL	Total estimated interest (INR)
maturity_amount	DECIMAL(18,2)	BigDecimal	NOT NULL	Total estimated maturity (INR)
estimated_maturity_date	DATE	LocalDate	NULLABLE	Estimated maturity date
created_at	DATETIME(3)	Instant	INDEX (idx_created_at)	Audit record creation timestamp

Key Audit Findings:

- Async Persistence: AuditService.saveAsync() is
- Failure Handling: If database is down, audit fail
- Simulation-Only: The table stores calculation es
created.
- Discrepancy: Documentation page 26 calls tabl

8. ARCHITECTURE BOUNDARY: FD CALCULATOR VS FD ACCOUNT MODULE

A fundamental architectural distinction exists between the Calculator and Account modules:

Lab	Module Responsibility	Classification	Calculator Impact
L6, L10, L11	FD Calculator (Simulation)	THIS SERVICE	Maturity estimation, product rate fetch, bonus math, CSV export.

Lab	Module Responsibility	Classification	Calculator Impact
L12, L13	Account Creation & Number Gen	FD ACCOUNT MODULE	fd_accounts table, branch-based account number generator. Out of scope.
L14	Interest Capitalization & Payout	FD ACCOUNT MODULE	Real account balance updates, daily interest accrual batch jobs. Out of scope.
L15, L16	Premature Withdrawal & Txn Log	FD ACCOUNT MODULE	Penalty deduction, fd_transactions initial deposit logging. Out of scope.
L17, L18	Batch Scheduler & Notifications	FD ACCOUNT MODULE	@Scheduled cron jobs, account status=CLOSED, SMS/email alerts. Out of scope.
L19	Maturity Payout & GL Transfer	FD ACCOUNT MODULE	Core banking ledger debit/credit, fund transfer to savings account. Out of scope.

9. REST API ENDPOINT SPECIFICATION

Method & URL	Auth	Role	Request Body	Response (200 OK) / Behavior
POST /api/v1/fd/calculations	JWT	CUSTOMER	CalculationRequest (principal, term, productCode)	CalculationResponse (effectiveRate, interestEarned, maturityAmount, explanation)
POST /api/v1/fd/calculations/compare	JWT	CUSTOMER	ComparisonRequest (principal, 2-3 product options)	ComparisonResponse (list sorted by maturityAmount descending)
GET /api/v1/fd/calculations/{id}	JWT	CUSTOMER	None	CalculationResponse (enforces ownership: returns 404 for non-owners)
GET /api/v1/fd/calculations/{id}/export	JWT	CUSTOMER	None (format=csv)	RFC 4180 CSV attachment (Content-Disposition header)
POST /api/v1/fd/calculations/{id}/explain	JWT	CUSTOMER	None	Plain-language English narrative explanation
GET /api/v1/fd/products/{code}/preview	JWT	CUSTOMER	None	ProductPreviewResponse (min/max bounds, rates, constraints)

10. AUTOMATED TESTING EVIDENCE (52/52 PASSING)

Test Class	Type	Count	Key Behaviors Verified
CalculationEngineImplTest.java	Unit	15	Simple & compound formulas, all 4 frequencies, Taylor series exponentiation, scale 2 rounding, large principals (9.99M), decimal values (1000.01).
CategoryBonusServiceImplTest.java	Unit	10	No categories, single category, multi-category sum, 2.0% cap enforcement, product max rate cap, unknown category handling, case normalization.
FdCalculatorServiceImplTest.java	Unit	13	Full orchestration, YEARS to MONTHS conversion, 404 missing product, 422 inactive product, 422 term bounds, 503 service outage, ownership checks.
FdCalculationControllerTest.java	Slice	6	401 on missing JWT, 403 on ADMIN role (no CUSTOMER), 400 on null principal, 400 on negative principal, 400 on missing productCode, 200 on valid.
FdCalculatorIntegrationTest.java	E2E	8	Real Spring context + WireMock stubs for Product & Customer services. End-to-end HTTP pipeline, bonus application, JWT security, Actuator health.

11. VIVA VOCE DEFENSE STRATEGY & RISK MANAGEMENT

High-Risk Questions (Where Gaps Exist):

1. "Slide 7 of the PPT says APY must be calculated. Where is APY calculated?"
   • Answer: "APY is not currently implemented in this version. The service computes the nominal effective rate and applies it to compounding. APY calculation = (1 + r/(100*n))^n - 1 is scheduled for the next release."

2. "Where is rate information cached until logout as specified in the PPT?"
   • Answer: "Rate caching is not yet implemented. In dev mode, MockProductClient provides zero network overhead. In production, we plan to implement Spring Cache with Caffeine/Redis with eviction hooked to the auth logout event."

3. "Can the user pass a custom start date for calculating maturity date?"
   • Answer: "Currently, CalculationRequest only accepts principal and term; the maturity date is computed from LocalDate.now(). Adding a startDate parameter to the request DTO is an immediate P0 task."

4. "Where is the Python reporting script mentioned in Lab L11?"
   • Answer: "We deliberately replaced the Python script with a native Java ExportService. Invoking Runtime.getRuntime().exec() creates severe command-injection risks and process overhead in web applications. Our native Java service exports RFC 4180 CSV directly."

Safe Questions (Core Strengths to Highlight):

• Why BigDecimal instead of double? Binary floating-point rounding errors (0.1 + 0.2 != 0.3) compound over millions of transactions. BigDecimal with HALF_UP rounding is the non-negotiable banking standard.
• How is data ownership enforced? Calculation retrieval queries by both calculationId AND the authenticated username from JWT. Accessing another customer's record returns HTTP 404 Not Found to prevent ID enumeration.

12. PRIORITIZED ENGINEERING FIX LIST

Priority	Item to Implement	Target Files	Implementation Details	Effort
P0 (Crucial)	Start Date input parameter	CalculationRequest.java, FdCalculatorServiceImpl.java	Add LocalDate startDate field to request DTO. Compute maturityDate = (startDate != null ? startDate : LocalDate.now()).plusMonths(termMonths).	30 min
P0 (Crucial)	APY Calculation	CalculationEngineImpl.java, CalculationResponse.java	Add APY formula (1 + r/(100*n))^n - 1 to engine and include BigDecimal apy in CalculationResponse.	45 min
P1 (High)	Rate Information Caching	pom.xml, ProductClient.java, CacheConfig.java	Add spring-boot-starter-cache + Caffeine. Annotate ProductClient.getCalculatorConfig with @Cacheable(value="productRates", key="#productCode").	1 hour
P1 (High)	Periodic Payout Math	CalculationEngineImpl.java, FdCalculatorServiceImpl.java	When payoutMode==PERIODIC, compute periodic payout amount = P * r / (100 * n) and return payout schedule.	1.5 hours
P2 (Medium)	Python Script Wrapper	scripts/generate_fd_report.py	Create a standalone Python script reading calculation ID and exporting CSV to show if professor demands Python.	30 min